

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change, Update]	138781	46.070829
1	[Git, Change, Create]	44929	14.914983
2	[Git, Change, Delete]	22781	7.562559
3	[Java, ByteCode, Member, Method]	13526	4.490197
4	[Java, ByteCode, Parameter]	13306	4.417164
5	[Git, Commit]	11390	3.781114
6	[File, Git]	8286	2.750686
7	[Java, ByteCode, ParameterizedType, Bound]	7286	2.418718
8	[Java, ByteCode, Bound]	7245	2.405107
9	[Git, Change, Rename]	7018	2.329750
10	[Java, ByteCode, Member, Field]	3611	1.198736
11	[Java, ByteCode, Bound, WildcardType]	2950	0.979305
12	[Java, Value, ByteCode, Annotation]	2936	0.974658
13	[Xml, Element]	2162	0.717714
14	[Java, ByteCode, Member, Constructor, Method]	2121	0.704104
15	[Xml, Text]	1450	0.481353
16	[Java, ByteCode, Bound, TypeVariable]	1111	0.368816
17	[Java, ByteCode, Member, Method, Lambda]	968	0.321345
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	890	0.295451
19	[Type, File, Java, Class, ByteCode]	845	0.280513
20	[Json, Key]	739	0.245324
21	[Value, Json, Scalar]	723	0.240013
22	[Java, Value, ByteCode, Primitive]	673	0.223414
23	[Type, File, Java, ByteCode, JavaType]	649	0.215447
24	[Java, ByteCode, Member, Method, GenericDeclar...	578	0.191877
25	[Type, File, Java, ByteCode, ExternalType]	412	0.136771
26	[Git, Change, Copy]	298	0.098926
27	[Author, Git, Person]	295	0.097931
28	[Value, Array]	289	0.095939
29	[Committer, Git, Person]	242	0.080336

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

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Nodes per label combination (more than 0.5% overall)

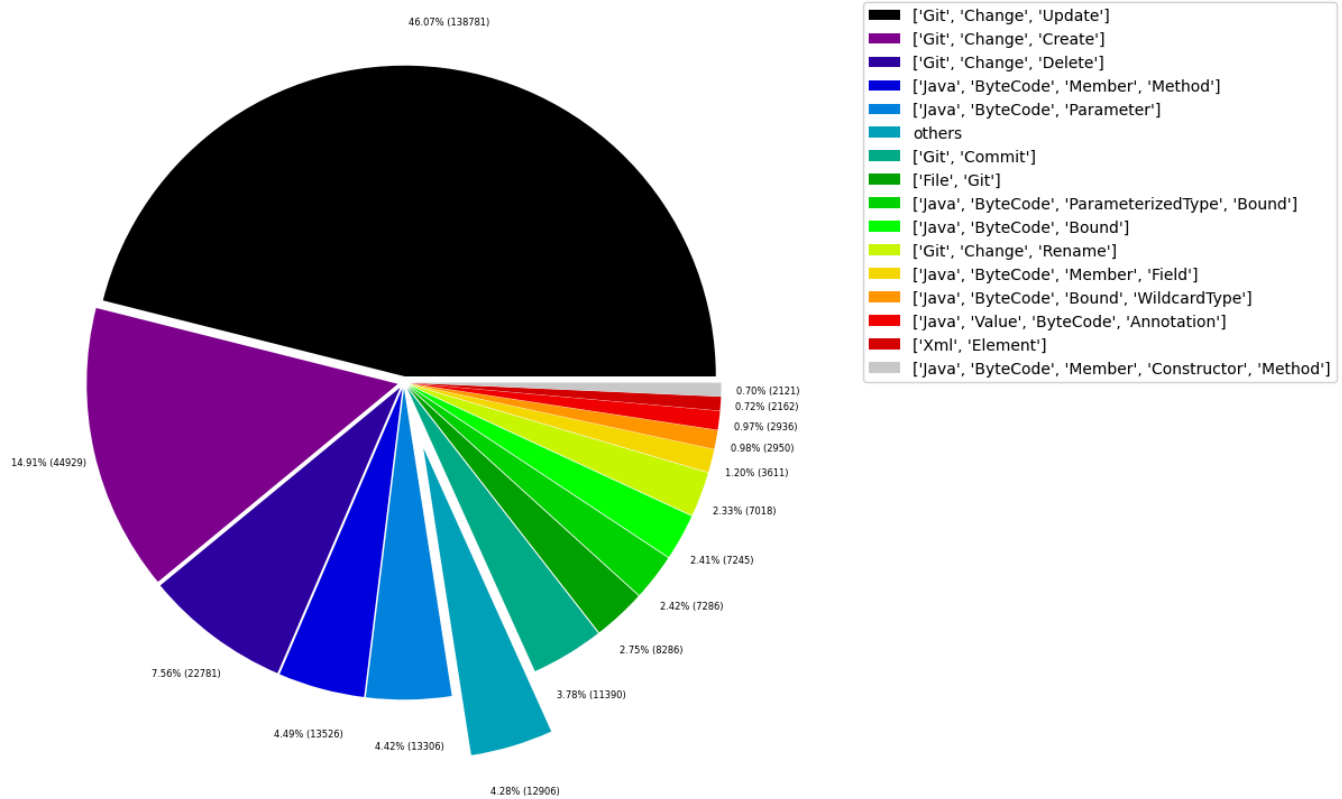


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000332
1	[Git, Branch]	1	0.000332
2	[Repository, File, Git]	1	0.000332
3	[File, Json]	2	0.000664
4	[File]	3	0.000996
5	[Java, ByteCode, Member, Constructor, Method, ...]	4	0.001328
6	[Maven, Exclusion]	5	0.001660
7	[Value, Json, Array]	6	0.001992
8	[File, Maven, Xml, Pom, Document]	9	0.002988
9	[Type, File, Java, ByteCode, Void]	9	0.002988
10	[Java, ManifestSection]	9	0.002988
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002988
12	[File, Java, Manifest]	9	0.002988
13	[File, Java, ServiceLoader]	10	0.003320
14	[File, Java, Properties]	12	0.003984
15	[Maven, ExecutionGoal]	16	0.005311
16	[Maven, PluginExecution]	16	0.005311
17	[Xml, Attribute]	18	0.005975
18	[jQAssistant, Rule, Concept]	19	0.006307
19	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006971
20	[Maven, Plugin]	21	0.006971
21	[Maven, Configuration]	21	0.006971
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007303
23	[Type, File, Java, ByteCode, Enum]	28	0.009295
24	[Type, File, Java, ByteCode, PrimitiveType]	30	0.009959
25	[Xml, Namespace]	36	0.011951
26	[Type, File, Java, ByteCode, Annotation]	44	0.014607
27	[File, Directory]	44	0.014607
28	[Type, File, Java, Class, ByteCode, Throwable]	55	0.018258
29	[Type, File, Java, ByteCode, Interface, Generi...	85	0.028217

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

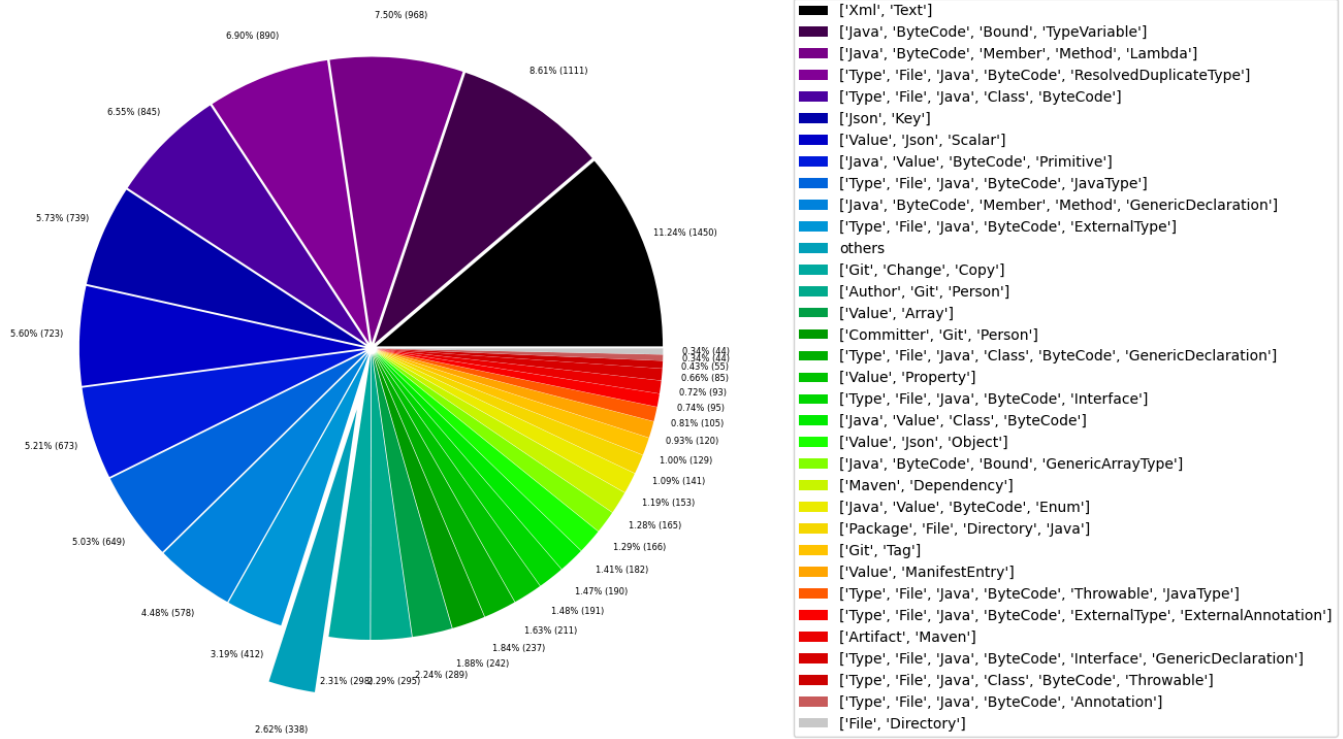


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

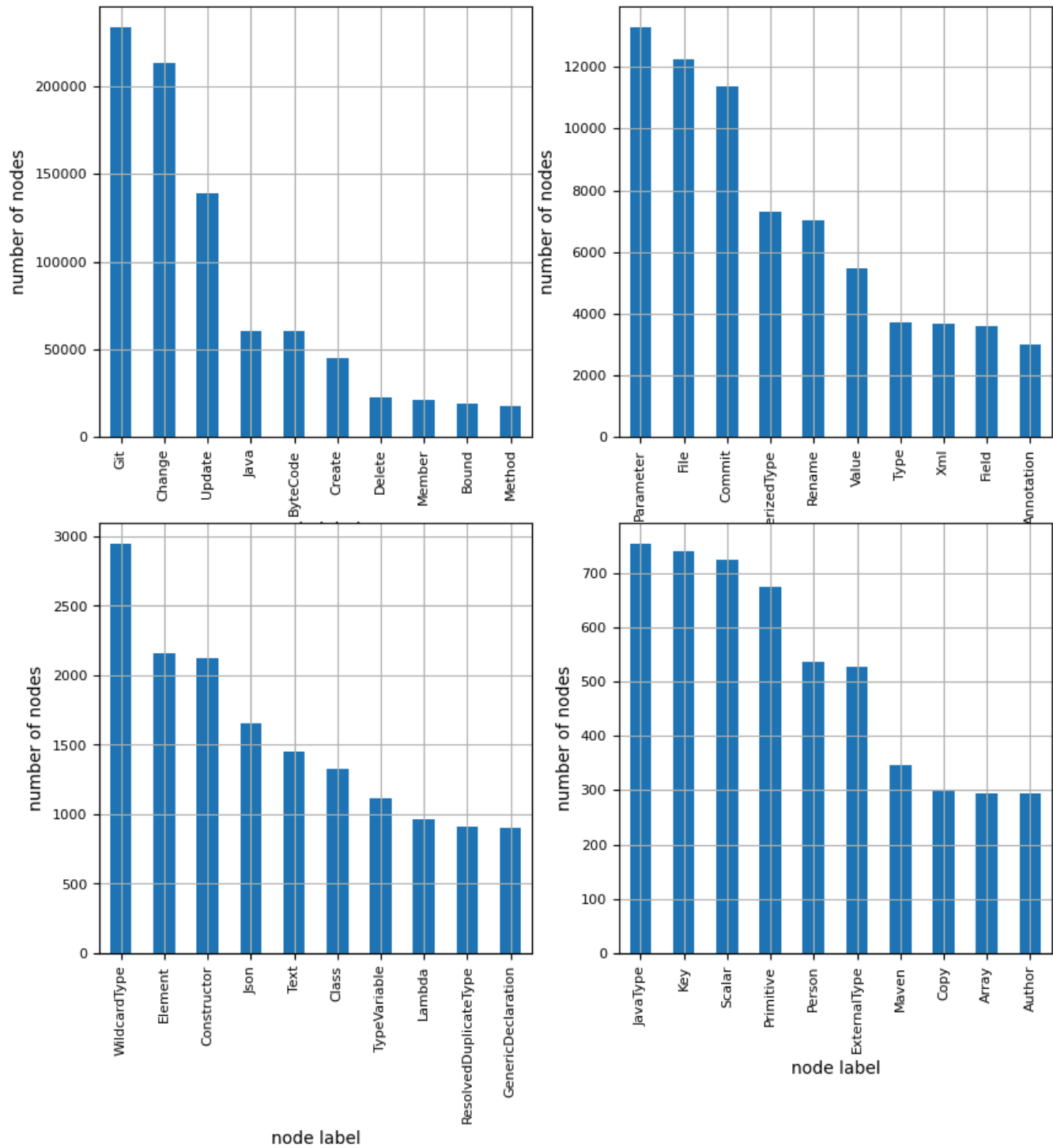
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	234151	77.730601
1	Change	213807	70.977048
2	Update	138781	46.070829
3	Java	60732	20.161071
4	ByteCode	60542	20.097997
5	Create	44929	14.914983
6	Delete	22781	7.562559
7	Member	20808	6.907587
8	Bound	18758	6.227053
9	Method	17197	5.708851
10	Parameter	13306	4.417164
11	File	12244	4.064614
12	Commit	11390	3.781114
13	ParameterizedType	7286	2.418718
14	Rename	7018	2.329750
15	Value	5483	1.820180
16	Type	3718	1.234256
17	Xml	3675	1.219982
18	Field	3611	1.198736
19	Annotation	2980	0.989264
20	WildcardType	2950	0.979305
21	Element	2162	0.717714
22	Constructor	2125	0.705432
23	Json	1652	0.548411
24	Text	1450	0.481353
25	Class	1327	0.440521
26	TypeVariable	1111	0.368816
27	Lambda	968	0.321345
28	ResolvedDuplicateType	912	0.302755
29	GenericDeclaration	904	0.300099
30	JavaType	754	0.250304
31	Key	739	0.245324
32	Scalar	723	0.240013
33	Primitive	673	0.223414
34	Person	537	0.178267
35	ExternalType	528	0.175279
36	Maven	346	0.114861
37	Copy	298	0.098926
38	Array	295	0.097931
39	Author	295	0.097931

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 935999

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	213807	22.842653
1	MODIFIES	213807	22.842653
2	UPDATES	138781	14.827046
3	CREATES	52245	5.581737
4	INVOKES	36643	3.914855
5	DELETES	29799	3.183657
6	COMMITTED	22780	2.433763
7	DEPENDS_ON	22504	2.404276
8	OF_TYPE	21945	2.344554
9	DECLARES	21268	2.272225
10	OF_RAW_TYPE	17367	1.855451
11	HAS	14444	1.543164
12	HAS_PARENT	13921	1.487288
13	RETURNS	12858	1.373719
14	HAS_COMMIT	11390	1.216882
15	READS	9382	1.002351
16	HAS_ACTUAL_TYPE_ARGUMENT	8414	0.898933
17	HAS_FILE	8286	0.885257
18	RENAMES	7018	0.749787
19	OF_GENERIC_TYPE	6000	0.641026
20	RESOLVES_TO	4241	0.453099
21	SIMILAR	4078	0.435684
22	WRITES	3948	0.421795
23	CONTAINS	3928	0.419659
24	HAS_NEW_NAME	3662	0.391240
25	RETURNS_GENERIC	3597	0.384295
26	CHANGED_TOGETHER_WITH	3034	0.324146
27	ANNOTATED_BY	2924	0.312393
28	REQUIRES	2233	0.238569
29	HAS_FIRST_CHILD	2162	0.230983

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

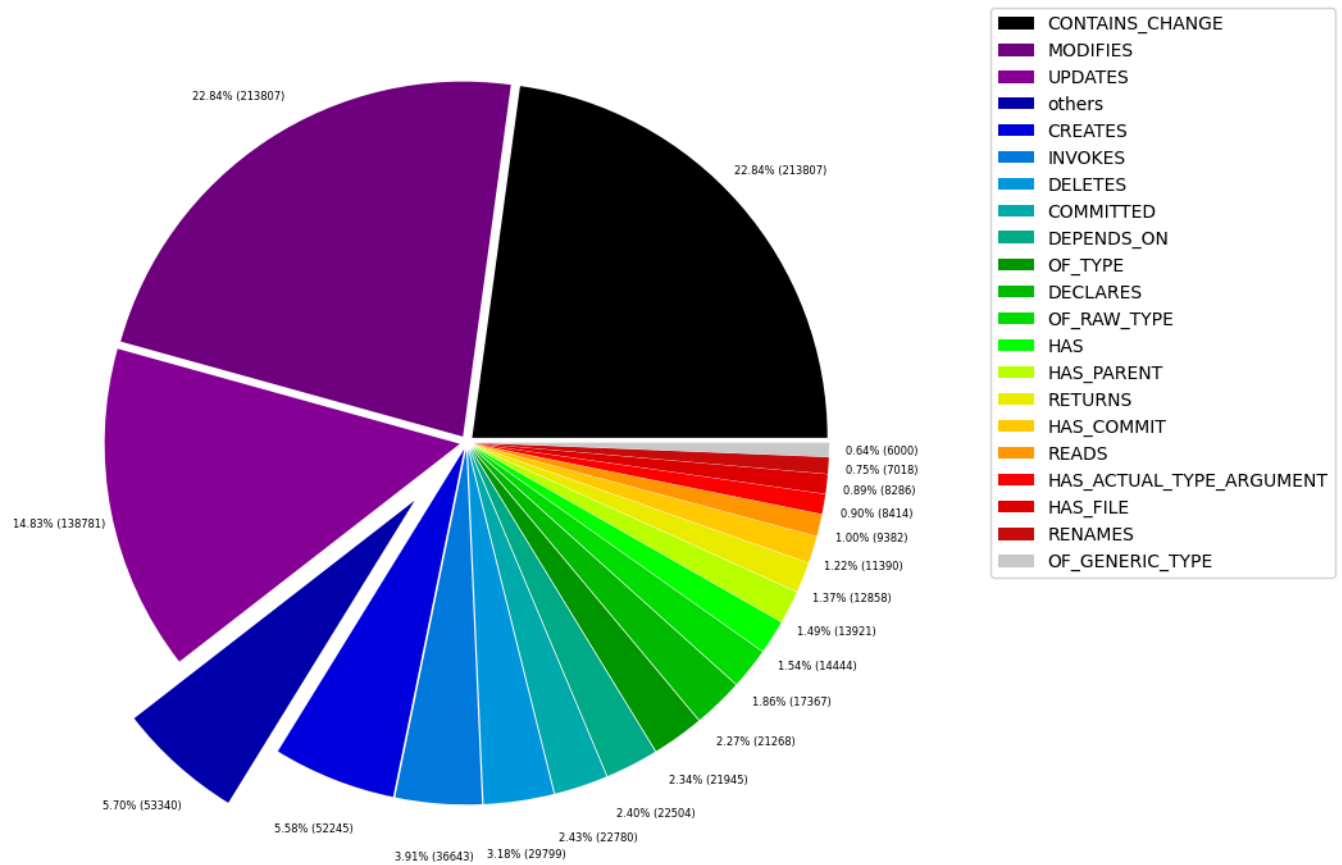


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

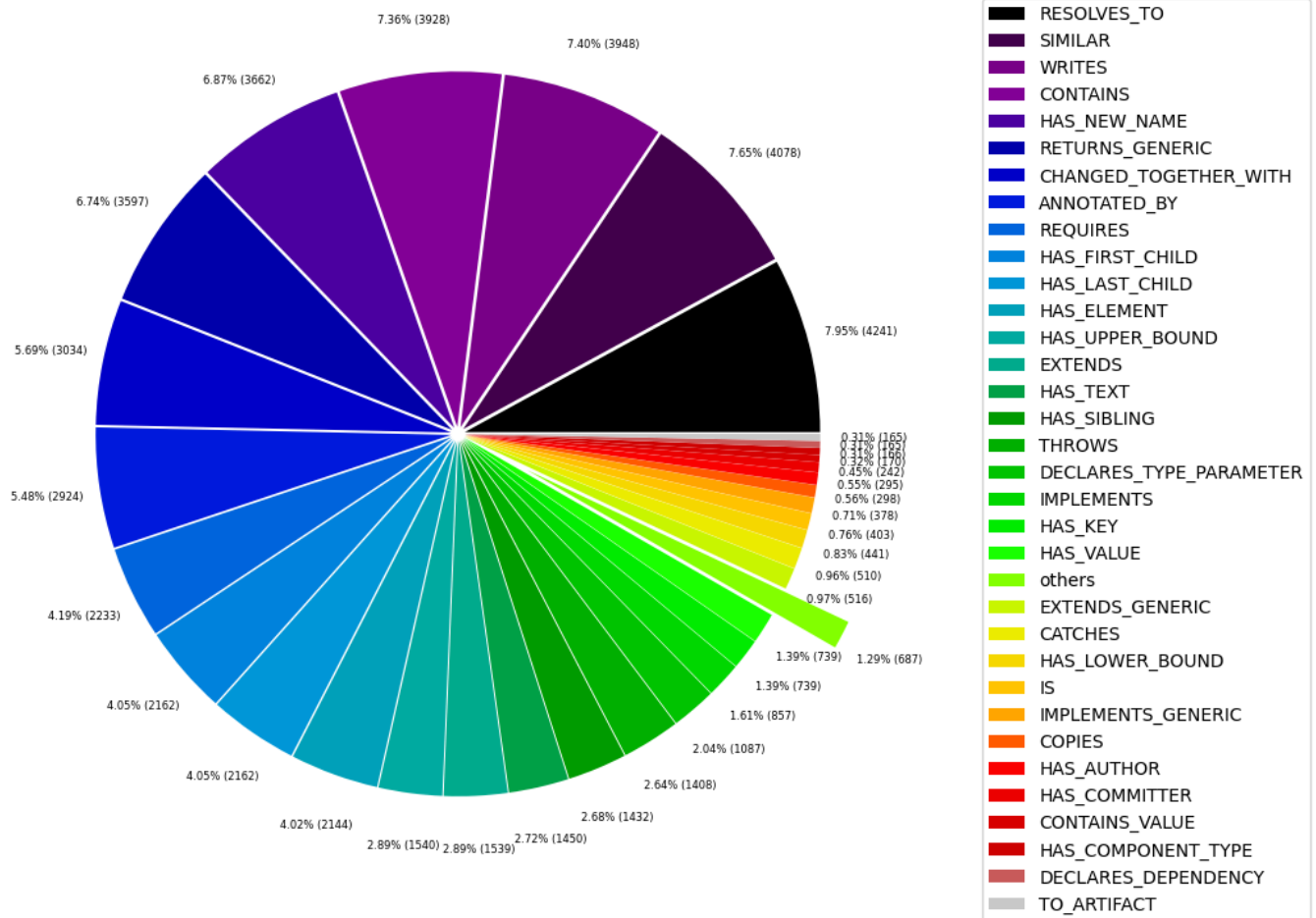
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000107
1	HAS_BRANCH	1	0.000107
2	HAS_HEAD	2	0.000214
3	THROWS_GENERIC	5	0.000534
4	EXCLUDES	5	0.000534
5	DESCRIBES	9	0.000962
6	HAS_ROOT_ELEMENT	10	0.001068
7	HAS_GOAL	16	0.001709
8	HAS_EXECUTION	16	0.001709
9	OF_NAMESPACE	18	0.001923
10	HAS_ATTRIBUTE	18	0.001923
11	INCLUDES_CONCEPT	19	0.002030
12	USES_PLUGIN	21	0.002244
13	IS_ARTIFACT	21	0.002244
14	HAS_CONFIGURATION	21	0.002244
15	REQUIRES_TYPE_PARAMETER	24	0.002564
16	REQUIRES_CONCEPT	28	0.002991
17	DECLARES_NAMESPACE	36	0.003846
18	HAS_DEFAULT	39	0.004167
19	COPY_OF	119	0.012714
20	HAS_TAG	129	0.013782
21	ON_COMMIT	129	0.013782
22	DECLARES_DEPENDENCY	165	0.017628
23	TO_ARTIFACT	165	0.017628
24	HAS_COMPONENT_TYPE	166	0.017735
25	CONTAINS_VALUE	170	0.018162
26	HAS_COMMITTER	242	0.025855
27	HAS_AUTHOR	295	0.031517
28	COPIES	298	0.031838
29	IMPLEMENTS_GENERIC	378	0.040385

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Change, Update]	MODIFIES	[File, Git]	138781	138781	
1	[Git, Change, Update]	UPDATES	[File, Git]	138781	138781	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Update]	138781	11390	
3	[Git, Change, Create]	MODIFIES	[File, Git]	44929	44929	
4	[Git, Change, Create]	CREATES	[File, Git]	44929	44929	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	44929	11390	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22781	11390	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	22781	22781	
8	[Git, Change, Delete]	DELETES	[File, Git]	22781	22781	
9	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22189	13426	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	13912	11390	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11390	1	
12	[Author, Git, Person]	COMMITTED	[Git, Commit]	11390	295	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11390	242	
14	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8515	13426	
15	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8375	13426	
16	[Repository, File, Git]	HAS_FILE	[File, Git]	8286	1	
17	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11390	
18	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
19	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
22	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6160	13306	
23	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8286	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3466	7245	
25	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3114	7286	
26	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7286	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, ParameterizedType, Bound]	2698	13306	
28	[Java, ByteCode, Member, Constructor, Method]	WRITES	[Java, ByteCode, Member, Field]	2520	2121	
29	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2475	7286	

Graph Density

```
total_number_of_nodes (vertices): 301234
total_number_of_relationships (edges): 935999
-> total directed graph density: 1.0314990897237496e-05
-> total directed graph density in percent: 0.0010314990897237497
```